

# Spatial Sampling Design and Covariance-Robust Minimax Prediction based on Convex Design Ideas

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**Abstract** This paper presents new ideas on sampling design and minimax prediction in a geostatistical model setting. Both presented methodologies are based on regression design ideas. For this reason the appendix of this paper gives an introduction to optimum Bayesian experimental design theory for linear regression models with uncorrelated errors. The presented methodologies and algorithms are then applied to the spatial setting of correlated random fields.

To be specific, in Section 1 we will approximate an isotropic random field by means of a regression model with a large number of regression functions with random amplitudes, similarly to Fedorov and Flanagan (1997). These authors make use of the Karhunen Loève approximation of the isotropic random field. We use the so-called polar spectral approximation instead; i.e. we approximate the isotropic random field by means of a regression model with sine-cosine-Bessel surface harmonics with random amplitudes and then, in accordance with Fedorov and Flanagan (1997), apply standard Bayesian experimental design algorithms to the resulting Bayesian regression model.

Section 2 deals with minimax prediction when the covariance function is known to vary in some set of a priori plausible covariance functions. Using a minimax theorem due to Sion (1958) we are able to formulate the minimax problem as being equivalent to an op-

timum experimental design problem, too. This makes the whole experimental design apparatus available for finding minimax kriging predictors. Furthermore some hints are given, how the approach to spatial sampling design with one a priori fixed covariance function may be extended by means of minimax kriging to a whole set of a priori plausible covariance functions such that the resulting designs are robust.

The theoretical developments are illustrated with two examples taken from radiological monitoring and soil science.

**Keywords** spatial sampling design · planning of monitoring networks · minimax prediction · kriging

## 1 Spatial sampling design

### 1.1 Introduction

A topic, getting so far not enough attention in the spatial statistics literature, is spatial sampling design, - the problem of optimal allocation of sampling points to spatial coordinates in order to improve spatial estimation and prediction in a well-defined sense. One reason for the neglect of model-based approaches to spatial sampling design might be the intractability of objective functions and corresponding optimization problems usually involved in spatial sampling design and the planning of monitoring networks. In contrast to non-spatial sampling design, we call it here experimental design, spatial sampling design becomes much more complicated, because spatial observations are correlated. The importance of (optimal) spatial sampling design considerations for environmental applications has been demonstrated in quite a few papers and monographs, we mention Brus and de Gruijter (1997), Diggle and

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Lophaven (2006) and Brus and Heuvelink (2007).

The experimental design problem for linear regression models with uncorrelated errors, however, may be considered as solved since the pioneering works of Kiefer (1959) and Fedorov (1972). The appendix gives an overview on the most important parts of experimental design theory and will be used extensively in the succession of this work.

In Section 1 we will approximate an isotropic random field by means of a regression model with a large number of regression functions with random amplitudes, similarly to Fedorov and Flanagan (1997), who make use of the Karhunen-Loeve approximation. In contrast to them we use the so-called polar spectral approximation and approximate the isotropic random field by means of a regression model with sine-cosine-Bessel surface harmonics with random amplitudes. Then, in accordance with Fedorov and Flanagan (1997), we apply standard Bayesian experimental design algorithms to the resulting Bayesian regression model.

Section 2 deals with minimax prediction when the covariance function is known to vary in some set of a priori plausible covariance functions. Using a minimax theorem due to Sion (1958) we are able to formulate the minimax problem as being equivalent to an optimum experimental design problem, too. This makes the whole experimental design apparatus available for finding minimax kriging predictors. Furthermore some hints are given, how the approach to spatial sampling design with one a priori fixed covariance function may be extended by means of minimax kriging to a whole set of a priori plausible covariance functions such that the resulting designs are robust.

The theoretical developments are illustrated with two examples taken from radiological monitoring and soil science.

## 1.2 Survey of model-based spatial sampling design

In this work a model-based approach to spatial sampling design will be followed. For a good general review on this topic we refer to Dobbie (2008), where also other approaches like stratified sampling, latin square sampling and randomized sampling are reviewed. Basic to the model-based approach is the fact that the spatial observations are modeled as realizations of a random field  $\{Y(x)|x \in \mathbf{X}\}$ , where  $\mathbf{X}$  is the spatial domain under investigation and  $Y(x)$  are random variables modeling the observations. Such random fields have the particular feature that different observations  $Y(x_1), Y(x_2)$  located

at different sites  $x_1, x_2 \in \mathbf{X}$  may be correlated. In the model-based approach to spatial sampling design this correlation is modeled, estimated and taken into account. Non-model-based sampling designs on the other hand are trying to get an efficient, space-filling sampling of the domain  $\mathbf{X}$ , often irrespective of the actual random field under investigation.

Often combined design criteria for simultaneous estimation and prediction are useful, too. But up to now in the literature either criteria for estimation of the covariance function or for prediction have been proposed, with only few exceptions: Zhu and Stein (2006) consider the minimization of the length of prediction intervals as design criterion and thereby combine both prediction and estimation of the covariance function. Apart from papers using the Bayesian paradigm and specifying priors also for covariance functions no other papers are known to the authors that combine estimation and prediction. The exceptions are Müller et al. (2004), Brown et al. (1994), Diggle and Lophaven (2006) and Fuentes et al. (2007).

Actually, Brown et al. (1994) and Fuentes et al. (2007) consider the covariance function to be non-stationary and deal with an entropy based design criterion according to which the determinant of the covariance matrix between locations to be added to the design must be maximized. Both make use of a simulated annealing algorithm to calculate the optimal designs. Müller et al. (2004) use a kind of simulated annealing algorithm, too, although their objective function has no explicit expression and must be found by means of an inhomogeneous Markov chain simulation. Diggle and Lophaven (2006) impressively demonstrate that designs taking account of both covariance function uncertainty and prediction accuracy must involve both samples very close to each other and space-filling samples on the domain of interest  $\mathbf{X}$ .

At this stage we are at a second distinguishing feature of spatial design algorithms developed in the literature so far. Most algorithms for finding optimal designs are heuristic stochastic search algorithms like simulated annealing (Aarts and Korst, 1989) that by no means fully exploit the mathematical structure of the formulated design criteria. The reasons might be that either the criteria are so complicated or too much time is necessary to exploit the mathematical structure of each single design criterion to get a result in reasonable time.

The present work attempts to fully exploit the mathematical structure of the design criteria at hand to

get fast deterministic design algorithms. It is based on works of Fedorov and Müller (1988), Fedorov and Hackl (1994) and Fedorov and Müller (2007) who use eigenfunction expansions of the random field under investigation to approximate it by means of a linear regression model with stochastic coefficients and then apply classical experimental design theory to this regression model. Further papers falling into this category, where it is tried to exploit classical experimental design theory for spatial sampling design, are Müller and Pazman (1998,1999), Pazman and Müller (2001), Müller and Pazman (2003) and Müller (2005). The first four papers deserve particular attention because there a new design measure and an approximate information matrix are investigated that has an interpretation as amount of added noise to design locations not considered to be important. Conventional experimental design algorithms for linear regression models with uncorrelated errors may be used to find optimal designs in this way. Further attention in this direction deserves the book by Melas (2006), who approximates designs by Taylor approximations and is able to monitor the robustness and change of designs by explicit formulae when parameters like the covariance function are changing.

Among the many papers that use the simulated annealing approach the following papers deserve special attention, either because of their special field of application, their design criteria, or their modeled covariance structure: Groenigen et al. (1999) is maybe the first manifestation of the catchword spatial simulated annealing (SSA). Trujillo-Ventura et al. (1991) consider multiobjective sampling design optimization. Brus and Heuvelink (2006) consider the minimisation of the average kriging variance, a design criterion that is considered also in this work under the heading of I-optimality, and apply simulated annealing to the design of a groundwater network. Finally, Müller et al. (2004), Brown et al. (1994) and Fuentes et al. (2007) have already been mentioned.

Works falling into none of the categories mentioned up to now are Bueso et al. (1999), Angulo and Bueso (2001), Chang et al. (2005) and Diggle et al. (2009): Bueso and Angulo (1999) present spatial sampling design, when the variable of interest cannot be directly observed, but information on it can be obtained by sampling a related variable. They formulate a stochastic-complexity-based criterion for spatial sampling design and make use of an EM algorithm to find optimal designs. Angulo and Bueso (2001) consider the same problem as Bueso and Angulo (1999), an unobservable variable. A method based on additive perturbation of the

variables of interest is proposed for the assignment of degrees of relative importance to the variables and/or locations in the design of sampling strategies. Chang et al. (2005) consider spatial sampling designs for extreme non-Gaussian phenomena. Diggle et al. (2009) investigate geostatistical analysis under preferential sampling by considering a log-Gaussian Cox process according to which the preferential samples are generated.

Today kriging and spatial sampling design find application not only in environmental sciences but also in the design of computer experiments. There explanatory variables entering the computer experiment as parameters are interpreted as spatial coordinates and the desired response surface is modeled as spatial random field over this high dimensional domain  $\mathbf{X}$  of parameters. Spatial sampling designs in computer experiments are directed towards accurate estimation of the response surface as well as towards identification of a certain point of the response surface, most often its maximum. The designs are therefore most often sequential. Prominent representatives in this direction are Mitchell and Morris (1992), Morris et al. (1993), Lim et al. (2002), Kleijnen (2004), Kleijnen and van Beers (2004) and Chen et al. (2006).

Apart from only one exception (Chang et al. 2005) the mentioned spatial sampling design papers all assume the random field to be Gaussian although the Bayesian methods mentioned could be easily adapted also to random fields with skew distributions gathered for example by means of a Box-Cox transformation of the variable of interest. Our own developments in the next sections implicitly make this Gaussian assumption, too. Future efforts should surely be directed towards generalization of the mentioned sampling design methods to skew and extreme value distributions since these are most often the distributions that are relevant in practice.

SERRA publications relevant to the present paper are Pilz and Spöck (2008), where the Bayesian approach to kriging is described in detail, and Recent Advances in Automatic Interpolation for Real-Time Mapping, SERRA Special Volume 22, Number 5, edited by Dubois (2008).

### 1.3 The spatial mixed linear model

This section will show that every isotropic random field can be approximated by a linear regression model with random coefficients. A spectral representation for isotropic random fields will be used to get this approximation. The approximation then allows the applica-

tion of powerful tools from convex optimization theory and the well-developed experimental design theory for Bayes linear regression models to calculate spatial sampling designs for random fields.

We consider a mean square continuous (m.s.c.) random field  $\{Y(x) : x \in \mathbf{X} \subseteq R^2\}$  such that

$$Y(x) = \mathbf{f}(x)^T \beta + \varepsilon(x), \quad E\varepsilon(x) = 0 \quad (1)$$

where  $\mathbf{f}(x)$  is a known vector of regression functions,  $\beta \in R^r$  a vector of unknown regression parameters and

$$\text{Cov}(Y(x), Y(y)) = C(x, y); \quad x, y \in \mathbf{X}. \quad (2)$$

It is well-known that, under the assumption of compactness of  $\mathbf{X}$  the error process of the m.s.c. random field has a uniformly convergent representation (in the m.s. sense)

$$\varepsilon(x) = \sum_{j=1}^{\infty} \sqrt{\lambda_j} \alpha_j \psi_j(x) \text{ for all } x \in \mathbf{X}, \quad (3)$$

where  $E(\alpha_j) = 0$ ,  $\text{Var}(\alpha_j) = 1$  and  $\text{Cov}(\alpha_i, \alpha_j) = 0$  for  $i \neq j$ ;  $i, j = 1, 2, \dots$ . The positive real numbers  $\lambda_j$  stand for the eigenvalues and the continuous real functions are the corresponding eigenfunctions of  $C(\cdot)$ , which then has a uniform and (absolute) convergent representation

$$C(x, y) = \sum_{j=1}^{\infty} \lambda_j \psi_j(x) \psi_j(y)$$

The components corresponding to the largest eigenvalues contribute most to the variability of the error process. Ordering the eigenvalues such that  $\lambda_1 \geq \lambda_2 \geq \dots$  and truncating the Karhunen-Loeve expansion at a sufficiently small eigenvalue  $\lambda_m$  we get an approximation of the random field (1) in the following form

$$Y(x) \approx \mathbf{f}(x)^T \beta + \sum_{j=1}^m \alpha_j g_j(x) + \varepsilon_0(x) \quad (4)$$

where  $g_j(x) = \sqrt{\lambda_j} \psi_j(x)$ ,  $E(\alpha_j) = 0$ ,  $\text{Cov}(\alpha_i, \alpha_j) = \delta_{ij}$ ;  $i, j = 1, \dots, m$  and it is assumed that the residual process  $\sum_{j=m+1}^{\infty} \sqrt{\lambda_j} \alpha_j \psi_j(x)$  can be sufficiently well approximated by a white-noise process with some variance  $\sigma_0^2$ . The right-hand side of (4) is exactly in the form of a mixed linear model, with fixed parameters  $\beta = (\beta_1, \dots, \beta_r)$  and random effects  $\alpha = (\alpha_1, \dots, \alpha_m)^T$ . This expansion of the covariance function was used by Fedorov and Flanagan (1997) and Fedorov and Müller (2007) in their work on spatial sampling design.

For practical applications, however, it will be very hard

to solve the generalized eigenvalue problem which makes the KL-expansion not very attractive as a starting point for spatial sampling design. When restricting attention solely to trend estimation, i.e. estimation of  $EY(x) = \mathbf{f}(x)^T \beta$ , then the results in Fedorov and Müller (2007) can be used for optimal design. It is, however, not obvious, how their approach can be applied to solve the more complicated design problem for prediction. We refer to Pilz and Spöck (2006), where the ideas following now have been presented for the first time.

#### 1.4 Polar Spectral Representation

From now on we will additionally assume that the m.s.c. random field is covariance-stationary and isotropic, i.e.

$$C(x, y) = C(\|x - y\|) \text{ for all } x, y \in \mathbf{X}.$$

Then, according to Yaglom (1987), the covariance function can be represented in the form

$$C(t) = \int_0^{\infty} J_0(t\omega) dG(\omega), \quad t \geq 0, \quad (5)$$

where  $J_0(\cdot)$  is the Bessel function of the first kind and order 0,  $t = \|x - y\|$  is the Euclidean distance between  $x$  and  $y$ , and  $G(\cdot)$  is the so-called (polar) spectral distribution function associated with  $C(\cdot)$ . As such  $G(\cdot)$  is positive, monotonically increasing and bounded from above. On the other hand, knowing  $C(\cdot)$  its spectral distribution can be obtained from the inversion formula

$$\frac{G(\omega^+) + G(\omega^-)}{2} = \int_0^{\infty} J_1(t\omega) \omega C(t) dt, \quad (6)$$

where  $G(\omega^+)$  and  $G(\omega^-)$  denote the right- and left-hand side limits at  $\omega$  and  $J_1$  denotes the Bessel function of first kind and order 1. Approximating  $G(\cdot)$  by means of a step function with positive jumps  $\delta_i = G(\omega_{i+1}) - G(\omega_i)$  at preselected points  $\omega_i$ ,  $i = 0, 1, \dots, n-1$ , and changing to polar coordinates  $(t, \varphi) = (\text{radius}, \text{angle})$  the polar spectral representation theorem for m.s.c. isotropic random fields tells us that the error process may be approximated as

$$\begin{aligned} \varepsilon(t, \varphi) \approx & \sum_{m=0}^{\infty} \{ \cos(m\varphi) \sum_{i=1}^n J_m(\omega_i t) U_{m,i} \} \\ & + \sum_{m=1}^{\infty} \{ \sin(m\varphi) \sum_{i=1}^n J_m(\omega_i t) V_{m,i} \}, \end{aligned}$$

where all the random variables  $U_{m,i}$  and  $V_{m,i}$  are uncorrelated, have mean zero, and their variances are  $\text{var}(U_{m,i}) = \text{var}(V_{m,i}) = d_m \delta_i$ ; and  $d_m = 1$  for  $m = 0$  and  $d_m = 2$  for  $m \geq 1$ . Again, by truncating the above

series at a sufficiently large  $m = M$ , we get an approximation of our random field in form of a mixed linear model

$$Y(x) \approx \mathbf{f}(x)^T \beta + \mathbf{g}(x)^T \alpha + \varepsilon_0(x) \quad (7)$$

The essential advantage over the previous

KL-representation is, however, that the above representation allows the explicit computation of the additional regression function  $\mathbf{g}(x)$  and the covariance matrix of the random vector  $\alpha$ . Noting that

$$\begin{aligned} \text{Cov}(\varepsilon(t_1, \varphi_1), \varepsilon(t_2, \varphi_2)) &\approx \\ &\approx \sum_{m=0}^M d_m \cos(m(\varphi_2 - \varphi_1)) \sum_{i=1}^n \delta_i J_m(t_1 \omega_i) J_m(t_2 \omega_i) \end{aligned}$$

it becomes clear that the components of the additional regression vector  $\mathbf{g}(\cdot)$  are made up of the following radial basis functions (cosine-sine-Bessel-harmonics)

$$\begin{aligned} g_{m,i}(t, \varphi) &= \cos(m\varphi) J_m(\omega_i t); \\ m &= 0, \dots, M; i = 1, \dots, n \\ g_{m,i}(t, \varphi) &= \sin((m - M)\varphi) J_{m-M}(\omega_i t); \\ m &= M + 1, \dots, 2M; i = 1, \dots, n \end{aligned}$$

### 1.5 Bayesian spatial linear model

Starting from our spatial mixed linear model (7) we may gain further flexibility with a Bayesian approach incorporating prior knowledge on the trend. To this we assume that the regression parameter vector  $\beta$  is random with

$$E(\beta) = \mu \in R^r, \quad \text{Cov}(\beta) = \Phi \quad (8)$$

This is exactly in the spirit of Omre (1987) who introduced Bayesian kriging this way. He used physical process knowledge to arrive at “qualified guesses” for the first and second order moments,  $\mu$  and  $\Phi$ . Some general ideas on specifying prior distributions for regression parameters may be found in Pilz (1991), ch. 3. On the other hand, the state of prior ignorance or non-informativity can be modelled by setting  $\mu = \mathbf{0}$  and letting  $\Phi^{-1}$  tend to the matrix of zeroes, thus passing the “Bayesian bridge” to universal kriging, see Omre and Halvorsen (1989).

Now, combining (7) and (8), we arrive at the **Bayesian spatial linear model** (BSLM)

$$Y(x) = \mathbf{h}(x)^T \gamma + \varepsilon_0(x) \quad (9)$$

where

$$\begin{aligned} \mathbf{h}(x) &= \begin{pmatrix} \mathbf{f}(x) \\ \mathbf{g}(x) \end{pmatrix}, \gamma = \begin{pmatrix} \beta \\ \alpha \end{pmatrix}, E\gamma = \begin{pmatrix} \mu \\ \mathbf{0} \end{pmatrix} =: \gamma_0, \\ \text{Cov}(\gamma) &= \begin{pmatrix} \Phi & \mathbf{0} \\ \mathbf{0} & \mathbf{A} \end{pmatrix} =: \Gamma \end{aligned}$$

Here  $\varepsilon_0(x)$  is white-noise with variance  $\sigma_0^2$  as before and  $\mathbf{A}$  denotes the covariance matrix of  $\alpha$ , resulting after the polar spectral approximation of the random field. Alternatively,  $\mathbf{g}$  and  $\mathbf{A}$  could also be chosen according to a truncated Karhunen-Loeve expansion.

Since in the approximated model (9) we have uncorrelated errors, the prediction of  $Y(x_0)$  at an unsampled location  $x_0 \in \mathbf{X}$  is equivalent to Bayes linear trend estimation,  $\hat{Y}(x_0) = \mathbf{h}(x_0)^T \hat{\gamma}_B$ . Thus, collecting our observations at given locations  $x_1, \dots, x_n \in \mathbf{X}$  in the vector of observations  $\mathbf{Y} = (Y(x_1), \dots, Y(x_n))^T$  and denot-

ing by  $\mathbf{H} = (h_i(x_j))^T = (\mathbf{F}; \mathbf{G})$ , where  $\mathbf{F} = (f_i(x_j))$ ,  $\mathbf{G} = (g_i(x_j))$  are the usual design matrices formed with the regression functions  $\mathbf{f}$  and  $\mathbf{g}$ , respectively, we obtain

$$\hat{Y}(x_0) = \mathbf{h}(x_0)^T (\mathbf{H}^T \mathbf{H} + \sigma_0^2 \Gamma^{-1})^{-1} (\mathbf{H}^T \mathbf{Y} + \sigma_0^2 \Gamma^{-1} \gamma_0)$$

Using the Sherman-Morrison-Woodbury matrix inversion formula the predictor  $\hat{Y}(x_0)$  may be rewritten as

$$\begin{aligned} \hat{Y}(x_0) &= \mathbf{f}(x_0)^T \mu + (\mathbf{G} \mathbf{A} \mathbf{g}(x_0) + \mathbf{F} \Phi \mathbf{f}(x_0))^T \\ &\quad [\sigma_0^2 \mathbf{I}_n + \mathbf{G} \mathbf{A} \mathbf{G}^T + \mathbf{F} \Phi \mathbf{F}^T]^{-1} (\mathbf{Y} - \mathbf{F} \mu) \end{aligned}$$

In this form, the analogy with the Bayesian kriging predictor

$$\begin{aligned} \hat{Y}_{BK}(x_0) &= \\ &= \mathbf{f}(x_0)^T \mu + (\mathbf{c}_0 + \mathbf{F} \Phi \mathbf{f}(x_0))^T [\mathbf{K} + \mathbf{F} \Phi \mathbf{F}^T]^{-1} (\mathbf{Y} - \mathbf{F} \mu) \end{aligned}$$

derived by Omre (1987) becomes obvious:  $\mathbf{G} \mathbf{A} \mathbf{g}(x_0)$  and  $\sigma_0^2 \mathbf{I}_n + \mathbf{G} \mathbf{A} \mathbf{G}^T$  approximate the vector of covariances

$\mathbf{c}_0^T = (\text{Cov}(Y(x_0), Y(x_1)), \dots, \text{Cov}(Y(x_0), Y(x_n)))$  and the covariance matrix  $\mathbf{K} = \text{Cov}(\mathbf{Y})$  of observations, respectively.

The total mean squared prediction error in the BSLM (9) is known to be

$$E(\hat{Y}(x_0) - Y(x_0))^2 = \quad (10)$$

$$= \sigma_0^2 (1 + \mathbf{h}(x_0)^T [\mathbf{H}^T \mathbf{H} + \sigma_0^2 \Gamma^{-1}]^{-1} \mathbf{h}(x_0)). \quad (11)$$

Using the same matrix inversion formula as above, the right-hand side of (11) may be written as

$$\begin{aligned} &\sigma_0^2 + \mathbf{g}(x_0)^T \mathbf{A} \mathbf{g}(x_0) + \mathbf{f}(x_0)^T \Phi \mathbf{f}(x_0) \\ &- (\mathbf{G} \mathbf{A} \mathbf{g}(x_0) + \mathbf{F} \Phi \mathbf{f}(x_0))^T [\sigma_0^2 \mathbf{I}_n + \mathbf{G} \mathbf{A} \mathbf{G}^T + \mathbf{F} \Phi \mathbf{F}^T]^{-1} \\ &\quad (\mathbf{G} \mathbf{A} \mathbf{g}(x_0) + \mathbf{F} \Phi \mathbf{f}(x_0)), \end{aligned}$$

which approximates the Bayesian kriging total mean squared error of prediction (TMSEP) derived by Omre (1987):

$$\begin{aligned} &E(\hat{Y}_{BK}(x_0) - Y(x_0))^2 = \\ &\sigma^2 + \mathbf{f}(x_0)^T \Phi \mathbf{f}(x_0) - (\mathbf{c}_0 + \mathbf{F} \Phi \mathbf{f}(x_0))^T [\mathbf{K} + \mathbf{F} \Phi \mathbf{F}^T]^{-1} \\ &\quad (\mathbf{c}_0 + \mathbf{F} \Phi \mathbf{f}(x_0)) \end{aligned}$$

along the same lines of reasoning as above. Thus, we have derived the important result that trend estimation in the approximating BSLM (9) is equivalent to Bayesian kriging in the original model.

### 1.6 Experimental design theory applied to spatial sampling design

The following section will apply the theory of optimal experimental design as discussed in the appendix to the spatial sampling design problem.

There are a number of well-defined design criteria according to which one might desire to optimally add or remove sampling locations  $x_1, x_2, \dots, x_n$  to or from a monitoring network. Besides design criteria involving both, maintenance costs of the monitoring stations and prediction accuracy, that can be solved only by means of the usage of time consuming heuristic search algorithms, like simulated annealing or genetic algorithms, the most important among these criteria certainly are the minimization of the integrated or the minimization of the maximum TMSEP over the area of investigation  $\mathbf{X}$ . In the following it will be shown that no computer intensive search heuristics are needed for the optimal calculation of such designs. On the contrary, the mathematical structure of these design criteria can be used to get improved fast algorithms for the finding of optimal designs. Two criteria will be considered: the I-optimality criterion

$$\int_{\mathbf{X}} E(\hat{Y}_{BK}(x_0) - Y(x_0))^2 dP(x_0) \rightarrow \min_{x_1, x_2, \dots, x_n}, \quad (12)$$

and a D-optimality criterion, for reasons that will become obvious later, when we make use of experimental design theory.  $P(x_0)$  in (12) is a probability measure on  $\mathbf{X}$  that has to be specified by the investigator, it is weighting different design points according to their importance in optimal design.

The last section has shown that the Bayesian kriging predictor and its TMSEP, both appearing in (12), are approximated by the Bayesian trend estimation of the BSLM (9) and the corresponding TMSEP (11). Replacing in (12) the TMSEP by its approximation (11) we arrive at design problems that are equivalent to standard Bayesian experimental design problems for the so-called I- and D-optimality criteria:

$$\int_{\mathbf{X}} \mathbf{h}(x_0)^T (\mathbf{H}^T(v_n) \mathbf{H}(v_n) + \sigma_0^2 \mathbf{\Gamma}^{-1})^{-1} \mathbf{h}(x_0) dP(x_0) \rightarrow \min_{v_n}$$

and

$$\det((\mathbf{H}^T(v_n) \mathbf{H}(v_n) + \sigma_0^2 \mathbf{\Gamma}^{-1})^{-1}) \rightarrow \min_{v_n}. \quad (13)$$

Here  $v_n = (x_1, x_2, \dots, x_n)$  collects the design points to be added or deleted from the monitoring network and  $\mathbf{H}(v_n)$  expresses the dependence of the design matrix  $\mathbf{H}$  on  $v_n$ .

At this point we advise the reader not familiar with Bayesian experimental design theory to read the appendix. The key point in this theory is that the above two so-called concrete design problems seemingly showing no mathematical structure may be expanded to so-called continuous design problems that have the nice feature to be convex optimization problems. Thus, the whole apparatus of convex optimization theory is available to approximately solve the above two design problems for I- and D-optimality. In particular, directional derivatives may be calculated and optimal continuous designs may be found by steepest descent algorithms. Continuous designs are just probability measures  $\xi$  on  $\mathbf{X}$  and may be rounded to exact designs  $v_n$ .

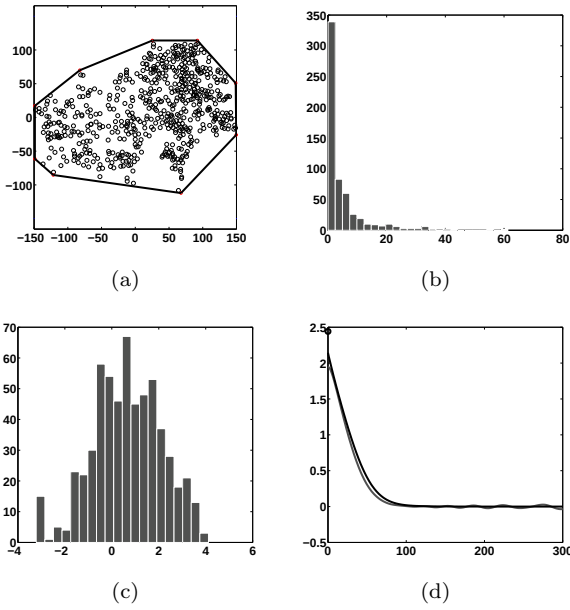
The next section is illustrating the foregoing theoretical developments by data examples and especially makes use of the exact design algorithms in the appendix. Although these algorithms do not guarantee to find the optimal exact design  $v_n$ , they give step-wise improvements of designs in such a way that the design functionals (12) and (13) decrease monotonically when design locations  $x_i$  are added. The closeness of the resulting exact designs  $v_n$  to the optimal continuous design  $\xi$  may be judged by means of a well-known efficiency formula (see, appendix), although we have found out that it is only a rough approximation to the true efficiency of the calculated designs. For this reason no efficiency calculations are given in the following illustration-section. Currently work is in progress that tries to make use of higher order Gateaux derivatives and of Taylor approximations to the design functionals. Efficiency estimates based on these approximations should be much better than those based only on first order Gateaux derivatives as described in the appendix. The corresponding calculations are not implemented, yet. We will report on this topic in a future paper.

One important note on D-optimality and the exact design algorithms from the appendix is appropriate here: According to these algorithms sampling locations with maximum TMSEP are added to the design in every step. Thus the approximation by a BSLM is not necessary at all, here, and these TMSEPs could be calculated also directly from the original covariance function.

## 1.7 Illustration

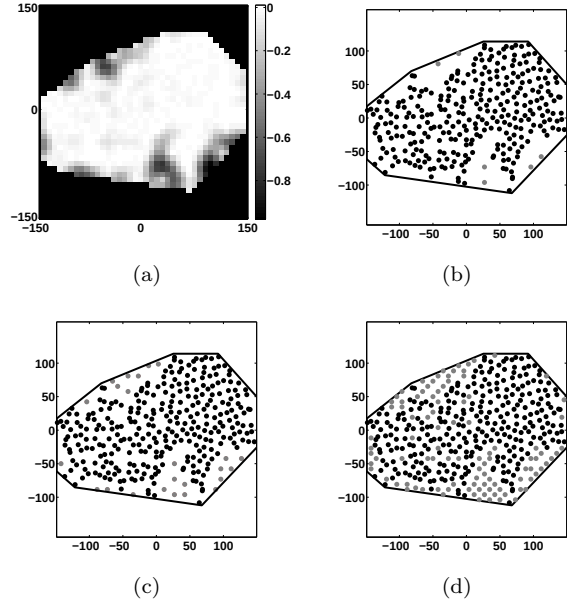
In what follows three illustrations for spatial sampling design are given. Two real data examples are considered that demonstrate the deletion and subsequent addition of design locations from and to a monitoring network. It will be shown that design locations can be saved and prediction accuracy can be improved by putting deleted monitoring stations at optimal new locations. The data considered are the Gomel- and the Jura data set. A third example is given to demonstrate the building of a monitoring network for prediction from scratch and to show results comparable to the literature. A fourth example demonstrates the robustness of optimal designs to the specification of the nugget effect.

### 1.7.1 Gomel data

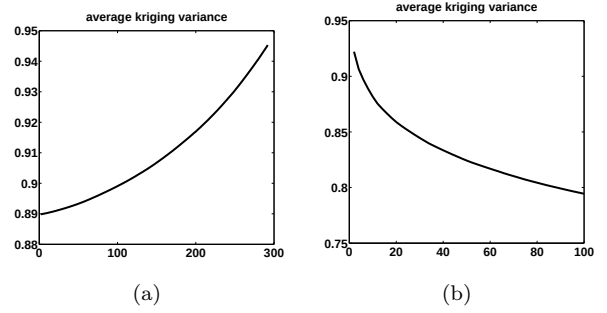


**Fig. 1** (a): 591 sample locations of the Gomel data. (b): Histogram of the Gomel data. (c): Histogram of the log-transformed Gomel data. (d): Covariance function of the log-transformed Gomel data and its worst case approximation.

The Gomel data are a set of 591 data locations in the Gomel district, Belarus, measuring Cesium-137 concentrations in the soil 10 years after the Chernobyl accident. Fig. 1 a) - d) show the 591 sample locations, their histogram, the histogram of the log-transformed concentrations and the corresponding isotropic covariance function that has been estimated by means of a weighted least squares fit of a convex combination of an exponential and Gaussian variogram model with



**Fig. 2** log-Gomel data. Ordinary kriging. (a): Directional derivative from 299+4 point design. (b): 299+6 point design. (c): 299+24 point design. (d): 299+100 point design.



**Fig. 3** log-Gomel data. Ordinary kriging. (a): Average kriging variance when locations are deleted. (b): Average kriging variance when locations are added.

different ranges to the empirical variogram of the log-transformed concentrations. It is visible from Fig. 1 a) that the sampling density is more dense in the East than in the West. A log-Gaussian isotropic random field for the transformed data may be justified by considering Fig. 1 c). Fig. 1 d) needs some clarification: - it displays not only the isotropic covariance function of the log-transformed data but also a approximation of this isotropic covariance function based on the polar spectral approximation of the random field for the log-transformed data by a BSLM:  $M = 35$  is selected as the largest angular frequency in the BSLM (9) and the radial frequencies are fixed at  $\omega_i = 0.001 * 1.125^{2i-1}$ ,  $i = 1, 2, \dots, 34$ . The origin of the coordinates is artificially set to the center of the considered region to get a good approximation of the whole random field and the least

favorable approximation to the covariance function is calculated along the horizontal line at 150 Northing. This way, assuming additionally a constant trend for the random field, a BSLM with 2415 regression functions results. As variance for the error of the BSLM  $\sigma_0^2 = 0.3 + 0.2$  is used, where 0.3 is the nugget effect of the true covariance function and 0.2 is recognized in Fig. 1 d) as the difference close to the origin  $t = 0$  between the true and the approximating covariance function. Thus, small scale variation in the true covariance function not taken into account by the approximating covariance function is modeled as a pure nugget effect.

Concerning the design problem, we were interested in I-optimality over a regular grid of points with grid-spacing equal to 3 and weighting measure  $P(\cdot)$  being constant. This approximation was used, because the integrals for calculating the matrix  $\mathbf{U}$  from the appendix in this way simplify to simple sums. Because we had no prior knowledge in this study and have assumed a constant trend we have fixed the prior mean  $\mu$  at 0 and selected a very large variance  $\Phi$ , such that all reported results correspond to ordinary kriging.

In a first stage, the sampling network has been reduced by means of the exact design algorithm 4.3.4 from the appendix to 299 stations remaining in the network. This reduced network is displayed in Fig. 2 b). Obviously most of the very close and redundant stations in the East have been removed. During the different iterations of the algorithm we have observed that the closest and most redundant stations are removed first. In the resulting design the remaining 299 stations are obviously equally spread and the design is very similar to a space-filling design. A look at Fig. 3 a) shows that obviously not much information has been lost by deleting 292 stations, because the average ordinary kriging variance calculated over the regular grid has only slightly increased from 0.89 to 0.945 after deleting the full network.

The reduced design with 299 stations has been used as a basis for the addition of further stations. In a different context of, for example, environmental monitoring of air pollution one could be interested in placing the saved expensive stations from the reduction step to other optimal locations. We made use of algorithm 4.3.3 from the appendix and added at every step two design points to the already available 299 sample locations. Sampling designs with  $299 + 4$ ,  $299 + 6$ ,  $299 + 24$  and  $299 + 100$  sampling locations are shown in Fig. 2 a) - d). Fig. 2 a) additionally shows directional derivatives from the optimal  $299 + 4$  point design. Light areas indicate the locations, where design points are already available; the

darkest areas indicate, where new design points should be selected. We propose to compare the darkest and lightest areas of Fig. 2 a) to Fig. 2 b), where two new design points are added. Fig. 3 b) shows the decrease in average kriging variance when new design points are optimally added to the available 299 sampling locations. The average kriging variance decreases from 0.945 for 299 sampling locations to 0.795 for  $299 + 100$  sampling locations. Already for 10 locations that are optimally added to the reduced network of 299 stations the new network of  $299 + 10$  stations has smaller average kriging variance than the original network of 591 stations although 290 stations have been saved.

At this stage some remarks on the properties of the resulting designs are appropriate:

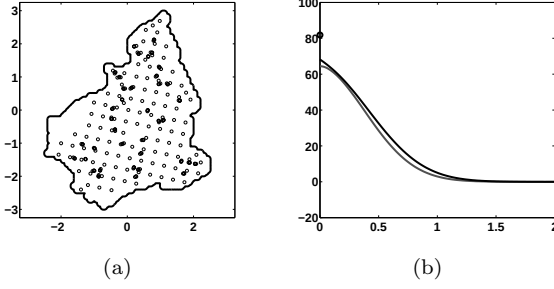
- Obviously additional design points have been selected at first in such areas which look empty: This property is reasonable because the kriging variance is most increased there.
- The first design points have been selected close to the boundary of the region of interest: This property is reasonable because the boundary gains information only from sampling locations on one side of the boundary.
- As a whole the resulting designs seem to be very similar to space filling designs: This is a property of the fact that the investigated design criterion is aimed at prediction and not at best estimation of the covariance function, where also sampling locations very close to each other would be needed to get, for example, the nugget effect properly estimated. A future paper will address also design for the estimation of the covariance function and a combined criterion for estimation and prediction with the main tools again borrowed from experimental design theory.
- Every design point has been selected only one time: This seems to be a feature of the polar spectral approximation but is not a property of the used design algorithm 4.3.3 that does not exclude repeated selection of design points.

### 1.7.2 Jura data

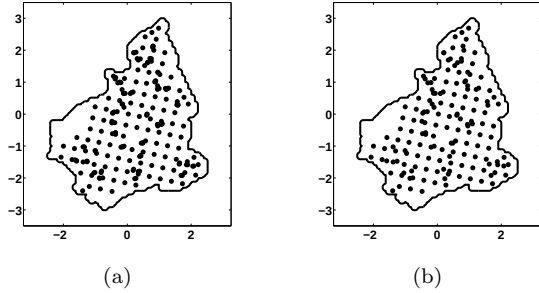
The Jura data set (Webster et al., 1994) contains 259 sampling locations with soil contamination measurements of trace metals in the Swiss Jura. It is a multivariate data set and was originally used to demonstrate coregionalization. In the following demonstration of spatial sampling design only the variable Ni is investigated. Fig. 4 shows the 259 sampling locations and the estimated covariance function for Ni with



its worst case approximation along the horizontal line with 3 Northing. The estimated covariance function is a convex combination of an exponential and a Gaussian covariance function model with different ranges. The approximating covariance function is again based on a BSLM with  $M = 35$  and the same frequencies  $\omega_i, i = 1, 2, \dots, 34$  as before, but scaled in such a way that  $\omega_{34} = 50$ . Again, having no prior knowledge on



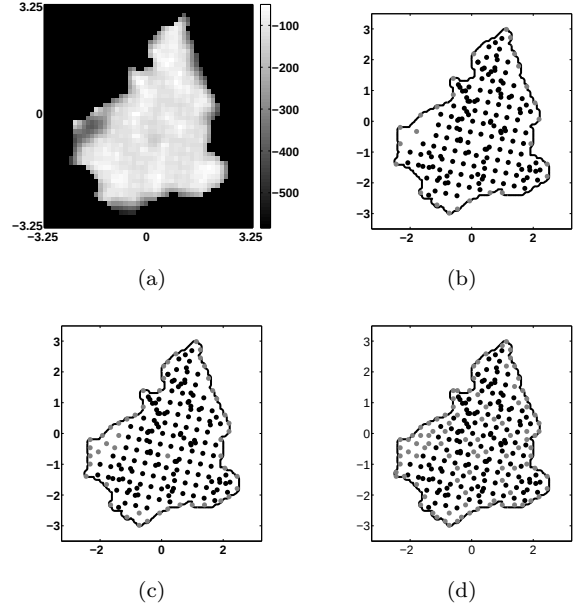
**Fig. 4** Jura data. (a): 259 sample locations. (b): Covariance function and its worst case approximation.



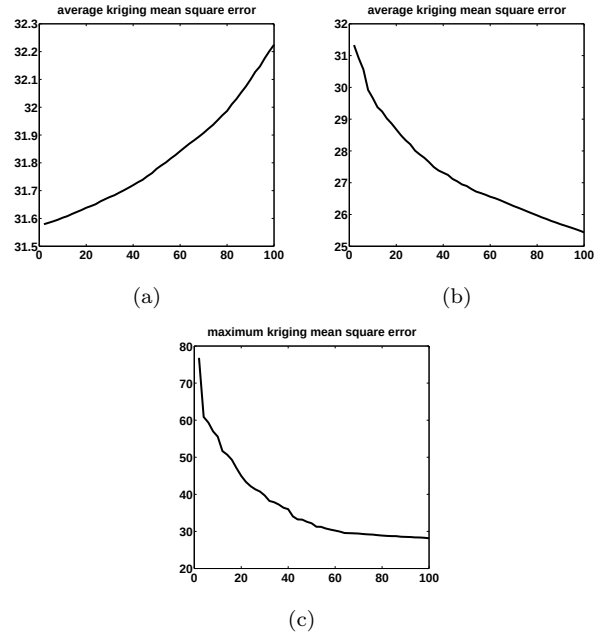
**Fig. 5** Jura data. (a): 50 samples deleted from design. (b): 100 samples deleted from design.

the mean of the isotropic random field the prior mean was set to  $\mu = 0$  along with a very large a priori variance  $\Phi$ . Thus, again, spatial sampling design for ordinary kriging is considered. But instead of the I-optimality criterion, as it was used for the Gornel data, the D-optimality criterion is considered here. The variance of the error of the BSLM was set to  $\sigma_0^2 = 13.6 + 3$ , where 13.6 is the nugget of the true covariance function and 3 is the difference between the true covariance function and its approximation close to lag  $t = 0$ .

As is visible from Fig. 4 the data set contains a lot of very close samples to estimate the covariance function appropriately near its origin. These are the first samples to get removed from the design, when algorithms 4.3.4 and 4.3.5 are applied (see Fig. 5). In this



**Fig. 6** Jura data. (a): Directional derivative from 159+2 point design. (b): 159+26 point design. (c): 159+50 point design. (d): 159+100 point design.



**Fig. 7** Jura data. Ordinary kriging. (a): Average kriging variance when locations are deleted. (b): Average kriging variance when locations are added. (c): Maximum kriging variance when locations are added.

way 100 samples have been removed from the design. Thereafter we have made use of algorithms 4.3.3 and 4.3.5 and have added again 100 locations to the remaining 159 sampling locations in a D-optimal way (Fig. 6). The resulting designs show good space filling properties and what has been said about the properties of I-

optimal designs in the last section is valid here, too, for the D-optimality criterion; although one property seems to be significant for the D-optimality criterion: Almost all of the first 26 design points have been selected directly at the boundary, where surely kriging variance is the largest. A look at Fig. 7 shows how the maximum kriging variance and the average kriging variance decrease, when design points are added in an optimal way according to the D-optimality criterion. The average kriging variance has been calculated over a regular grid of points with grid spacing 0.06.

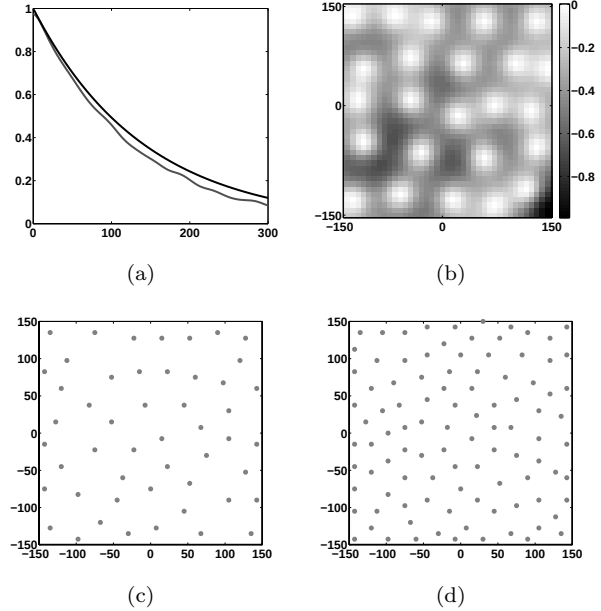
### 1.7.3 Starting from scratch

The aim of this third example is to demonstrate the building of a monitoring network for prediction from scratch. The region investigated is a square area with length 300. It is assumed from a pre-study or from a-priori knowledge that the covariance function may be fixed as an exponential one with nugget 0, sill 1 and range  $\sqrt{2} * 300$ . The purpose of the investigation is to generate I-optimal designs for prediction without the availability of any starting locations to which new design locations should be added.

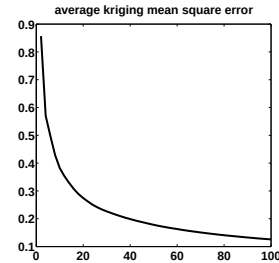
Similarly as before it has been decided to locate the origin of the coordinate system at the center of the square area to get good approximations of the complete region by means of a BSLM (9). The frequencies of the BSLM again are fixed at  $M = 35$  and the same frequencies  $\omega_i, i = 1, 2, \dots, 34$  are used as in example 1.7.1 but scaled in such a way that the largest frequency is  $\omega_{34} = 1.65$ .

Fig. 8 a) shows the covariance function and its worst case approximation at the horizontal line with 150 Northing. The difference between the two covariance functions at lag  $t = 0$  is 0.025 and this is exactly the value that was used for  $\sigma_0^2$  in the BSLM. In Fig. 8 b) - d) I-optimal designs calculated by means of the exact design algorithm 4.3.3 from the appendix are visualized. The designs show good coverage of the complete square area and are very similar to space-filling designs although the distance between adjacent design points is not uniform. Notably, the closer design points are to the border of the square area the smaller is the distance to the closest adjacent design point. This feature may be explained from the fact that the border area gets information only from one side of the border whereas design points close to the center of the square region get information from their complete surrounding. Fig. 9 shows the decrease in average ordinary kriging variance, when design locations

are added. The average kriging variance is calculated on a square grid with grid spacing 3.



**Fig. 8** Ordinary kriging. (a): Covariance function and its worst case approximation. (b): Directional derivative from 24 point design. (c): 50 point design. (d): 100 point design.

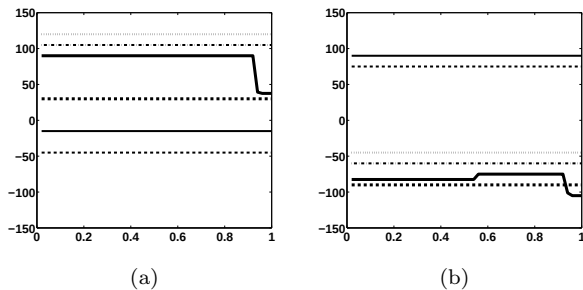


**Fig. 9** Ordinary kriging. Average kriging variance when locations are added.

### 1.7.4 Robustness against the choice of $\sigma_0^2$

The BSLM (9) is crucially dependent on the variance  $\sigma_0^2$  for the uncorrelated error process  $\epsilon_0(\cdot)$ . Small scale variation of the true model not taken into account by the BSLM (9) is modeled by  $\epsilon_0(\cdot)$  and its variance  $\sigma_0^2$ . In the previous examples the true covariance function was approximated by the one from the BSLM and  $\sigma_0^2$  was specified as the difference between the true covariance function and its approximation close to lag  $t = 0$  plus the nugget effect from the true model. The intention of this last computational example is to investigate

the robustness of the optimal designs against the specification of  $\sigma_0^2$ . The Gomel data and 6-point I-optimal designs are considered. For that purpose  $\sigma_0^2$  has been



**Fig. 10** Gomel data. Ordinary kriging. Robustness of 6-point design to the specification of  $\sigma_0^2$ . (a): Easting coordinates. (b): Northing coordinates.

varied in 49 steps between 0.02 and 1 and I-optimal  $299 + 6$  - point designs have been calculated by means of algorithm 4.3.3 from the appendix. Fig. 10 shows the Easting and Northing coordinates of the 6 added design locations for the resulting I-optimal designs at the y-axes. On the x-axes  $\sigma_0^2$  is varied.  $\sigma_0^2 = 0.5$  corresponds to the value used for the calculation of the designs in example 1.7.1. Obviously the designs are quite robust against the specification of  $\sigma_0^2$ , change only 3 times and then only at one design location. Results for designs with more than 6 added design locations are similar.

Robustness of designs is also an issue in the last chapter on minimax kriging. There the robustness of designs will be considered not only against the specification of  $\sigma_0^2$  but also against the specification of the complete covariance function model.

## 2 Covariance-Robust Minimax Kriging

### 2.1 Introduction

In chapter 1 an approach to spatial sampling design for the Bayesian kriging predictor has been elaborated. For the calculation of spatial sampling designs we made use of Bayesian experimental design theory. This second part of the paper will exploit this theory, too, albeit in a first instance for the prediction problem and then in the second instance also for spatial sampling design related to this prediction problem.

The spatial prediction method of kriging is based on the assumption that the covariance function is exactly known. In practical applications this assumption does

not hold, because the covariance function is just an estimate. As a consequence the true prediction error in kriging is underestimated (Christensen, 1991), and also the praised BLUP-optimality of kriging does no longer hold. The dramatic consequences of neglecting the effects of an underestimation of the actual MSE of kriging has been further detailed by Stein (1999). He points to the fact that up to now there is no satisfactory frequentist solution to correcting the bias of estimated covariance parameters and robust evaluation of EBLUPs.

The uncertainty of the covariance estimation will be taken into account here by means of a frequentist approach. Our approach to specifying uncertainty makes use of the so-called minimax principle. Applying this principle, we want to find a predictor in such a way that the maximum possible risk of this predictor becomes a minimum among all a priori possible and plausible covariance functions. Thus, minimax prediction protects against the worst possible scenario. Our covariance-robust minimax predictor provides a solution to the problem of robustifying EBLUPs. It's relevance to environmental applications is obvious. The need for robust and reliable predictions of critical environmental variables has been demonstrated, not the least, in the recent special SERRA volume edited by Dubois (2008). The minimax predictor also has a Bayesian justification, since it will turn out as a Bayesian kriging predictor with respect to the least favourable a priori distribution.

Minimax kriging has been proposed for the first time in Pilz et al. (1997) although no complete and detailed derivation of this predictor and also no application to data was given there. The present paper gives the first detailed mathematical derivation, calculation and application of this predictor to a real data problem. More importantly, the last subsection of this paper also gives some hints for the calculation of spatial sampling designs for this predictor. These designs are robust against the specification of the complete covariance function. This and the fact that for the calculation of the minimax kriging predictor extensive use of experimental design theory is made are the reasons why a discussion of this predictor also fits to the first part of the paper, where spatial sampling design within the context of experimental design is discussed.

In Sections 2.2 and 2.3 we formulate the minimax problem and specify the minimax predictor as least favourable Bayesian kriging predictor. Next in Section 2.4 we show the equivalence of the minimax problem to a Bayesian experimental design problem. Algorithms known from

Bayesian experimental design theory as discussed in the appendix may be used to calculate the minimax predictor. In this context the explicit form of a directional derivative is given that may be used to calculate the minimax predictor by means of steepest descent algorithms. We close the chapter by presenting numerical examples in Section 2.5. Especially Section 2.5.2 is of importance because it demonstrates how spatial sampling designs being robust against the specification of the covariance function may be calculated.

## 2.2 The model

The model for covariance-robust minimax kriging is similar to the model for Bayesian linear kriging, with the exception that the covariance function is assumed to be unknown and to stem from a parametrically specified class  $\mathcal{C}$  of a priori plausible covariance functions. The covariance functions  $C_\theta(x_1, x_2)$  may have, for example, the parametric form of Matern covariance functions with parameter domain  $\Theta$  assumed to be compact. Thus, the model for minimax kriging reads

$$\begin{aligned} Y(x) &= \mathbf{f}(x)^T \beta + \epsilon(x), \quad x \in \mathbf{X} \subset \mathbb{R}^m \\ E(Y(x)|\beta, \theta) &= \mathbf{f}(x)^T \beta \\ \text{cov}(Y(x_1), Y(x_2)|\beta, \theta) &= C_\theta(x_1, x_2) \in \mathcal{C}, \\ \theta &\in \Theta \text{ compact} \\ E(\beta|\theta) &= \mu \\ \text{cov}(\beta|\theta) &= \Phi. \end{aligned}$$

## 2.3 Characterization of the minimax predictor

In order to take account of the uncertainty of the covariance function specified in the above model we will try to determine a predictor  $\hat{Y}$  for the prediction of unknown values  $Y(x_0)$  in the class of affine linear predictors  $D_a = \{\hat{Y} = \mathbf{w}^T \mathbf{Y} + w_0; \mathbf{w} \in \mathbb{R}^n, w_0 \in \mathbb{R}\}$  in such a way, that the maximum possible total mean squared error of prediction (TMSEP) among all a priori plausible covariance functions  $C_\theta(x_1, x_2) \in \mathcal{C}, \theta \in \Theta$ , becomes a minimum. We will call this predictor covariance-robust minimax kriging predictor.

The TMSEP of a predictor  $\hat{Y} \in D_a$  corresponding to some a priori plausible covariance function  $C_\theta \in \mathcal{C}$  is given by

$$\begin{aligned} \text{TMSEP}(C_\theta; \hat{Y}) &= E_\beta E_{\mathbf{Y}_0|\beta, \theta}((\hat{Y} - Y(x_0))^2) = \\ &= C_\theta(x_0, x_0) + \mathbf{f}(x_0)^T \Phi \mathbf{f}(x_0) - 2\mathbf{w}^T (\mathbf{c}_{0, \theta} + \mathbf{F} \Phi \mathbf{f}(x_0)) \\ &\quad + \mathbf{w}^T (\mathbf{K}_\theta + \mathbf{F} \Phi \mathbf{F}^T) \mathbf{w} + \{\mu^T (\mathbf{f}(x_0) - \mathbf{F}^T \mathbf{w}) - w_0\}^2, \end{aligned}$$

where  $\mathbf{c}_{0, \theta}$  specifies the covariance between the point to be predicted and the observed data  $Y(x_1), \dots, Y(x_n)$ ,

$\mathbf{K}_\theta$  is the covariance matrix of the observed data and  $\mathbf{F} = (\mathbf{f}(x_1), \dots, \mathbf{f}(x_n))^T$  is the design matrix. Since in the squared marginal bias  $\{\mu^T (\mathbf{f}(x_0) - \mathbf{F}^T \mathbf{w}) - w_0\}^2$  the covariance function does not appear, the covariance-robust minimax kriging predictor  $\hat{Y}^M(x_0)$  may be sought in the class of all marginally unbiased predictors  $D = \{\hat{Y} = \mathbf{w}^T \mathbf{Y} + \mu^T (\mathbf{f}(x_0) - \mathbf{F}^T \mathbf{w}) | \mathbf{w} \in \mathbb{R}^n\}$  and is characterized by the relationship

$$\sup_{C_\theta \in \mathcal{C}} \text{TMSEP}(C_\theta; \hat{Y}^M(x_0)) = \inf_{\hat{Y} \in D} \sup_{C_\theta \in \mathcal{C}} \text{TMSEP}(C_\theta; \hat{Y})$$

This definition of the minimax kriging predictor necessitates that the infima and suprema exist. We therefore suppose that the parametrization of the covariance function is continuous on the compact set  $\Theta$ , ie.  $\lim_{\theta \rightarrow \theta_0} C_\theta(x_1, x_2) = C_{\theta_0}(x_1, x_2)$  for all  $x_1, x_2 \in \mathbf{X}$ . Under this assumption  $\text{TMSEP}(C_\theta; \hat{Y})$ , interpreted as a function of  $\theta$ , is a continuous function with compact domain  $\Theta$ , for every fixed  $\hat{Y} \in D$ . Lemma 6.5 of Pilz (1991) then shows that

$$\sup_{C_\theta \in \mathcal{C}} \text{TMSEP}(C_\theta; \hat{Y}) = \sup_{\xi \in \Xi} \int_{\Theta} \text{TMSEP}(C_\theta; \hat{Y}) \xi(d\theta),$$

where  $\Xi$  is the set of all probability measures  $\xi$  defined on the  $\sigma$ -algebra of the Borel sets of  $\Theta$ .

The minimax problem

$$\begin{aligned} \sup_{C_\theta \in \mathcal{C}} \text{TMSEP}(C_\theta; \hat{Y}^M(x_0)) &= \\ \inf_{\hat{Y} \in D} \sup_{\xi \in \Xi} \int_{\Theta} \text{TMSEP}(C_\theta; \hat{Y}) \xi(d\theta), \end{aligned}$$

which, as a consequence of the last two equations is equivalent to the original minimax problem, may now be solved by interchanging minimization and maximization. This follows from a minimax theorem of Sion (1958) and the fact that the set of so-called average covariance matrices

$$\begin{aligned} \mathbf{M}(\xi) &= \\ &= \int_{\Theta} \begin{pmatrix} C_\theta(x_0, x_0) & \mathbf{c}_{0, \theta}^T \\ \mathbf{c}_{0, \theta} & \mathbf{K}_\theta \end{pmatrix} \xi(d\theta) =: \begin{pmatrix} C_\xi(x_0, x_0) & \mathbf{c}_{0, \xi}^T \\ \mathbf{c}_{0, \xi} & \mathbf{K}_\xi \end{pmatrix}, \\ &\quad \xi \in \Xi \end{aligned}$$

is convex and compact. The convexity and compactness of the set of average covariance matrices  $\mathbf{M}(\xi)$  follows from the continuity of the covariance functions  $C_\theta(x_1, x_2)$  with respect to  $\theta$  directly from Lemma 5.1.8 in Bandemer et al. (1977) by interpreting the average covariance matrices as information matrices known from experimental design theory. For a proof of this fact see also Spöck (1997). The minimax theorem of Sion (1958) may be applied since

$$\begin{aligned} \int_{\Theta} \text{TMSEP}(C_\theta; \hat{Y}) \xi(d\theta) &= \\ &= \mathbf{C}_\xi(x_0, x_0) + \mathbf{f}(x_0)^T \Phi \mathbf{f}(x_0) \\ &\quad - 2\mathbf{w}^T (\mathbf{c}_{0, \xi} + \mathbf{F} \Phi \mathbf{f}(x_0)) + \mathbf{w}^T (\mathbf{K}_\xi + \mathbf{F} \Phi \mathbf{F}^T) \mathbf{w} \end{aligned}$$

is a continuous and concave function in  $\mathbf{M}(\xi)$  and a convex function in  $\mathbf{w} \in \mathbb{R}^n$ . The average covariance matrix  $\mathbf{M}(\xi)$  sought in the minimax problem and the weight vector  $\mathbf{w} \in \mathbb{R}^n$  are saddle points of the above integrated TMSEP, and this is the reason why the minimax theorem of Sion works.

Interchanging supremum and infimum we see that the integrated TMSEP becomes a minimum if and only if we insert for  $\hat{Y}$  the Bayesian kriging predictor

$$\hat{Y}_{\mathbf{M}(\xi)}^B(x_0) = \mathbf{f}(x_0)^T \boldsymbol{\mu} + (\mathbf{c}_{0,\xi} + \mathbf{F}\boldsymbol{\Phi}\mathbf{f}(x_0))^T (\mathbf{K}_\xi + \mathbf{F}\boldsymbol{\Phi}\mathbf{F}^T)^{-1} (\mathbf{Y} - \mathbf{F}\boldsymbol{\mu}),$$

where  $\mathbf{Y} = (Y(x_1), \dots, Y(x_n))^T$  is the data vector. The minimax predictor thus is characterized as the Bayesian kriging predictor  $\hat{Y}_{\mathbf{M}(\xi_0)}^B$  that maximizes the Bayes risk

$$\text{TMSEP}(\mathbf{M}(\xi); \hat{Y}_{\mathbf{M}(\xi)}^B) = \quad (14)$$

$$= C_\xi(x_0, x_0) + \mathbf{f}(x_0)^T \boldsymbol{\Phi}\mathbf{f}(x_0) - (\mathbf{c}_{0,\xi} + \mathbf{F}\boldsymbol{\Phi}\mathbf{f}(x_0))^T (\mathbf{K}_\xi + \mathbf{F}\boldsymbol{\Phi}\mathbf{F}^T)^{-1} (\mathbf{c}_{0,\xi} + \mathbf{F}\boldsymbol{\Phi}\mathbf{f}(x_0)) \quad (15)$$

w.r.t.  $\xi \in \Xi$ . That is, the covariance-robust minimax kriging predictor is characterized as the least favourable Bayesian kriging predictor.

#### 2.4 The equivalence of the minimax problem to a convex design problem

Instead of maximizing the functional (14) with respect to  $\mathbf{M}(\xi)$  the equivalent problem of minimization of the reciprocal of (14) is considered. The block-inversion rule for regular matrices allows one to identify this minimization problem with a  $c_b$ -optimal Bayesian design problem known from experimental design theory. Applying this inversion rule to the so-called average total covariance matrix

$$\mathbf{M}_b(\xi) = \begin{pmatrix} C_\xi(x_0, x_0) + \mathbf{f}(x_0)^T \boldsymbol{\Phi}\mathbf{f}(x_0) & (\mathbf{c}_{0,\xi} + \mathbf{F}\boldsymbol{\Phi}\mathbf{f}(x_0))^T \\ \mathbf{c}_{0,\xi} + \mathbf{F}\boldsymbol{\Phi}\mathbf{f}(x_0) & \mathbf{K}_\xi + \mathbf{F}\boldsymbol{\Phi}\mathbf{F}^T \end{pmatrix},$$

the first block in the inverse  $\mathbf{M}_b(\xi)^{-1}$  is given by

$$\text{TMSEP}(\mathbf{M}(\xi); \hat{Y}_{\mathbf{M}(\xi)}^B)^{-1}, \quad (16)$$

the reciprocal of the integrated Bayesian kriging TMSEP. Inversion of  $\mathbf{M}_b(\xi)$  is possible, since we assume the covariance functions to be positiv definite, and  $\mathbf{M}_b(\Xi) = \{\mathbf{M}_b(\xi) | \xi \in \Xi\}$  is the convex hull of all total covariance matrices. Multiplication of the matrix  $\mathbf{M}_b(\xi)^{-1}$  with the unit vector  $\mathbf{c}_b = (1, 0, 0, \dots, 0)^T \in \mathbb{R}^{n+1}$  in the quadratic form  $\mathbf{c}_b^T \mathbf{M}_b(\xi)^{-1} \mathbf{c}_b$  thus results exactly in the expression (16) to be minimized.

Defining

$$\mathbf{U}_b = \mathbf{c}_b \mathbf{c}_b^T$$

and

$$\Psi(\xi) = \Psi(\mathbf{M}_b(\xi)) = \text{tr}(\mathbf{U}_b \mathbf{M}_b(\xi)^{-1})$$

with  $\text{tr}(\cdot)$  the trace functional, the minimax predictor may thus be determined by minimizing the functional  $\Psi(\cdot)$  with respect to  $\mathbf{M}_b(\xi)$ .

Before minimizing this functional some of its properties by analogy with information matrices and design functionals from experimental design theory should be collected:

- First of all it easily follows from the compactness and convexity of the set  $\mathbf{M}(\Xi)$  that the set  $\mathbf{M}_b(\Xi)$  is compact and convex, too.
- Analogously to the proof of Theorem 12.2 in Pilz (1991) it may be shown that the above functional is convex on the set  $\mathbf{M}_b(\Xi)$ . Furthermore it may be shown that the functional  $\Psi(\mathbf{M}_b(\xi))$  is continuous with respect to the usual sum of squares metric on  $\mathbf{M}_b(\Xi)$ .
- Lemma 11.6 and 11.7 in Pilz (1991) show that for all pairs  $(\xi, \bar{\xi}) \in \Xi \times \Xi$  the directional derivatives

$$\Delta_\Psi(\xi, \bar{\xi}) = \lim_{\alpha \downarrow 0} \frac{\Psi((1-\alpha)\xi + \alpha\bar{\xi}) - \Psi(\xi)}{\alpha}$$

exist and are given by

$$\Delta_\Psi(\xi, \bar{\xi}) = \quad (17)$$

$$= \Psi(\xi) - \text{tr}(\mathbf{U}_b \mathbf{M}_b(\xi)^{-1} \mathbf{M}_b(\bar{\xi}) \mathbf{M}_b(\xi)^{-1}). \quad (18)$$

- It may be shown that the directional derivative attains its minimum in the direction of a probability measure  $\xi_\theta$  with one-point support  $\theta \in \Theta$ . Thus

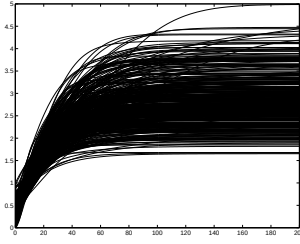
$$\inf_{\bar{\xi} \in \Xi} \Delta_\Psi(\xi, \bar{\xi}) = \inf_{\theta \in \Theta} \Delta_\Psi(\xi, \bar{\xi}_\theta).$$

The directional derivative plays a prominent role in the construction of algorithms for the minimization of the convex functional  $\Psi(\cdot)$ . As a consequence of the analogy of this functional to design functionals known from experimental design theory, gradient descent algorithms may be used as they are known from this theory. At this stage we refer to the appendix 4.2, where one such minimization algorithm due to Fedorov (1972) is proposed for the minimization of design functionals. The same algorithm may be used for the minimization of the convex functional  $\Psi(\cdot)$ , too. Merely the directional derivative in this optimum design algorithm has to be replaced by the expression (17).

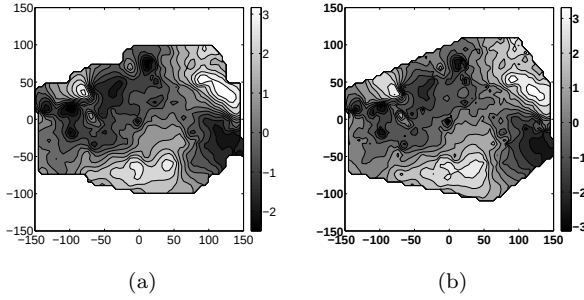
#### 2.5 Illustration

##### 2.5.1 Minimax kriging of log-transformed Gomel data

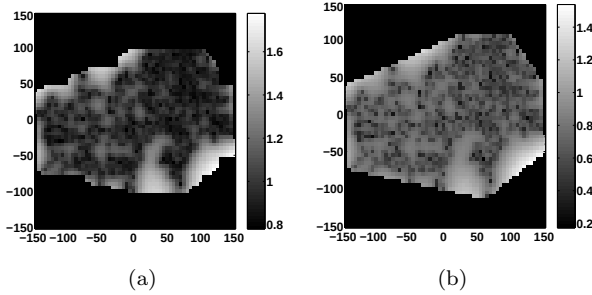
The data discussed in this illustration of covariance-robust minimax kriging are the log-transformed Gomel



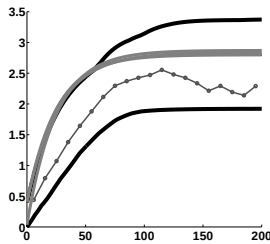
**Fig. 11** log-Gomel data. Belt of bootstrapped (simulated) semivariograms.



**Fig. 12** log-Gomel data. (a): Ordinary minimax kriging predictions. (b): Ordinary kriging predictions.



**Fig. 13** log-Gomel data. (a): Square roots of the Ordinary minimax kriging risks. (b): Ordinary kriging standard deviations.



**Fig. 14** log-Gomel data. Belt of plausible semivariograms (dark), semivariograms of the minimax predictors (light grey) and empirical semivariogram (grey).

data of section 1.7.1. Ordinary minimax kriging of this log-transformed data is considered; i.e. the a priori mean  $\mu$  of the constant trend is set to 0 and is given a very large a priori variance  $\Phi$ .

At first a semivariogram was estimated from the log-transformed data by means of an empirical semivariogram estimator and then a theoretical semivariogram model was fitted to this empirical semivariogram by means of weighted least squares. The theoretical semivariogram model to be fitted was a convex combination of an exponential and a Gaussian semivariogram model with different ranges. To get some insight into the uncertainty of the semivariogram estimate a parametric bootstrap has been used: At the original 591 sampling locations Gaussian random fields have been simulated, with a theoretical semivariogram model equal to the semivariogram that has been estimated from the original data by weighted least squares; and for every Gaussian simulation the semivariogram has been re-estimated by means of weighted least squares. For re-estimation a convex combination of an exponential and a Gaussian semivariogram model with different ranges has been used, again. Fig. 11 illustrates 500 semivariograms that have been parametrically bootstrapped in this way. The belt of bootstrapped semivariograms may be regarded as almost 100% confidence region for the true but unknown semivariogram. Obviously the uncertainty is largest for the nugget effect and the sill of the true semivariogram. The gradient of the semivariogram close to the origin  $t = 0$  seems to be estimated quite well.

For minimax kriging a compact parameter domain  $\Theta$  of semivariogram parameters  $\theta = (\text{nugget, partial sill, exponential range, Gaussian range, mixing parameter})$  is needed: For this purpose after taking a look at Fig. 11 the parameter domain has been restricted first by  $(0, 1.75, 75, 75, 0) \leq \theta \leq (0.4, 3, 150, 150, 1)$ ; secondly, by restricting this set of parameters by a further constraint on the Mahalanobis distance of the plausible semivariograms from the semivariogram estimated for the log-transformed Gomel data by means of weighted least squares. As weighting matrix we used the one obtained from weighted least squares of the original log-transformed data. Using visual inspection it was found out that all semivariograms having Mahalanobis distance smaller than 7 times the Mahalanobis distance of the empirical semivariogram estimate and its weighted least squares estimate, fitted well the uncertainty represented in the semivariogram simulations. Fig. 14 shows the belt of semivariograms produced by the above two restrictions.

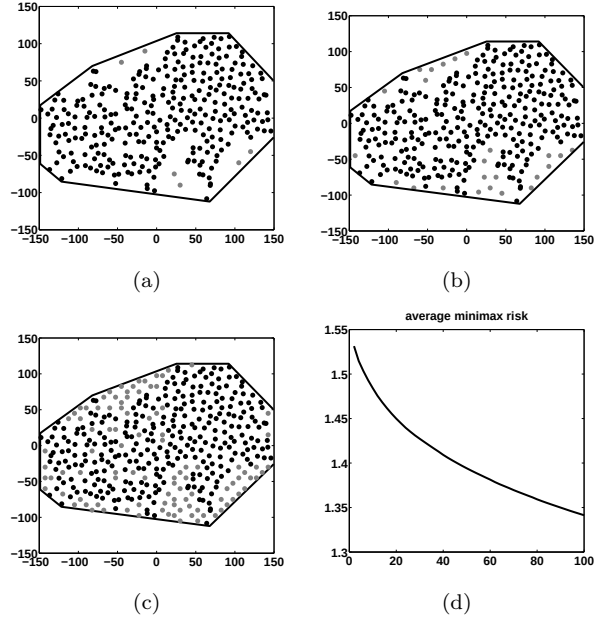
The ordinary minimax kriging predictors  $\hat{Y}_{\mathbf{M}(\xi)}^B(x_0)$  have been calculated on a square grid of  $60 \times 60$  spatial locations, where the same matrix  $\mathbf{M}_b(\xi)$  has been used for all 9 locations surrounding a point  $x_0$  on the square

grid. The calculations took about 24 hours on a PC with 3GB RAM and a 3GHz processor. For the calculations we used algorithm 4.2 from the appendix and they were run until an efficiency of 99% was reached. Fig. 12 shows the prediction results for ordinary minimax kriging and for ordinary kriging which has been based on the estimated semivariogram. Comparing both figures we see that the minimax kriging predictor and the ordinary kriging predictor give similar results. Fig. 13 gives the corresponding square roots of the minimax risks and the ordinary kriging standard deviations, which actually are always smaller than the square roots of the minimax risks because the minimax predictor is a least favourable ordinary kriging predictor. Fig. 14 shows all the semivariograms at which the minimax predictors are realized as least favourable ordinary kriging predictors. Notably, and this fact is important, as we will see in the next section, these semivariograms are almost identical and are placed at the upper boundary of the belt of a priori plausible semivariograms: It seems that the minimax kriging predictor or least favourable ordinary kriging predictor is attained at a semivariogram model with maximum possible nugget, least possible range and average sill. That is, minimax kriging seems to be kriging with the random field assumed to be as rough as plausible based on prior knowledge. Actually this is a hint how minimax predictors can be calculated approximately in practice without making use of the complicated algorithms discussed in this paper: After specifying, e.g. by means of bootstrapping semivariogram estimates as discussed in this paper, the most rough representation (largest nugget, smallest range, average sill) of the random field under consideration just use ordinary kriging with this semivariogram model and your kriging predictor becomes least favourable and thus a minimax predictor.

### 2.5.2 Robust spatial sampling design

In the following spatial sampling design for the covariance-robust minimax kriging predictor will be discussed. Sampling designs for the minimax kriging predictor are robust to the specification of the covariance function because this predictor is designed in such a way that it minimizes the maximum possible TMSEP over a complete class of a priori plausible covariance functions.

The key to spatial sampling design for the minimax kriging predictor is visualized in Fig 14. There all the semivariograms are visualized at which the minimax kriging predictors are realized as least favourable ordinary kriging predictors. Apart from small differences due to their iterative calculation by means of algorithm



**Fig. 15** log-Gomel data. Ordinary minimax kriging. (a): 299+6 point design. (b): 299+24 point design. (c): 299+100 point design. (d): Average minimax risk when locations are added.

4.2 from the appendix and a terminating criterion based on 99% efficiency these semivariograms are all the same for all locations  $x_0$  on the  $60 \times 60$  square grid where prediction was envisaged. That is, minimax kriging over the complete area of investigation  $\mathbf{X}$  corresponds to one least favourable ordinary kriging predictor, whose unique semivariogram is illustrated in Fig. 14.

Spatial sampling design for the minimax kriging predictor thus is equivalent to spatial sampling design for the mentioned least favourable ordinary kriging predictor with unique semivariogram. Among the only slightly differing light grey semivariograms of Fig. 14 we have selected the one with maximum nugget, maximum sill and smallest range and have approximated the corresponding isotropic random field by means of a BSLM (9). Again a worst case approximation to the covariance function was calculated and  $\sigma_0^2$  in the BSLM (9) was selected as the difference between the just discussed covariance function and its approximation close to lag  $t = 0$ . As design criterion the I-optimality criterion was used. Fig. 15 shows some designs based on this criterion and also the decrease in average minimax risk, when locations are added to the already available 299 sampling locations. Fig. 15 should be compared to Fig. 2 and Fig. 3: Obviously the designs have changed, but the space-filling property has been retained. The designs in Fig. 15 have the additional advantage that they are robust

in the minimax-sense to a complete class of a priori plausible covariance functions.

### 3 Conclusion

The preceding chapters have discussed spatial sampling design for prediction and its connection with experimental design for Bayesian linear regression models. By means of approximating an isotropic random field by its polar spectral representation consisting of a superposition of cosine-sine-Bessel surface harmonics it was possible to formulate the spatial design problem as being equivalent to an experimental design problem for Bayesian linear regression. In this way powerful convex optimization tools from this theory can be used to calculate spatial sampling designs. Reduction of environmental monitoring networks and addition of new sampling locations to networks has been considered. Two design criteria, I- and D-optimality have been discussed and illustrated by examples.

We note that our approach is not restricted to isotropic random fields only, geometrically anisotropic random fields can be approached in the same way: Such random fields simply have to be transformed to isotropic ones by means of a linear transformation of the coordinate system. Currently work is in progress to extend the approach proposed in this paper also to non-stationary random fields, which locally maintain isotropy (stationarity) called locally isotropic (stationary) random fields. For more details on these we refer to Spöck (2008).

Covariance-robust minimax kriging has been proposed because of its robustness against a complete class of a priori plausible covariance functions, its computational equivalence to a Bayesian experimental design problem, but mainly because of the fact that spatial sampling design for this predictor makes the resulting designs robust against not knowing the exact covariance function. This fact is relevant in practice, especially when too few samples are available to get a reliable covariance function estimate. For this case we have proposed to parametrically bootstrap the covariance estimate to get an insight into its uncertainty and to define the belt of plausible covariance functions for minimax kriging or spatial sampling design. Furthermore we have argued that a Bayesian kriging predictor whose covariance function has maximum possible nugget, maximum sill, smallest range and steepest descent close to the origin might be considered least favourable and therefore minimax. To be robust in the minimax sense our proposal therefore is to calculate spatial sampling designs

always for a Bayesian kriging predictor whose covariance function a priori has maximum plausible nugget, maximum plausible sill, smallest plausible range and steepest plausible descent close to the origin. Currently work is making progress to justify this empirical observation from illustrations 2.5.1 and 2.5.2 by means of a closer look at the polar spectral densities of the a priori plausible covariance functions.

As indicated in Section 1.6 a future paper will give also efficiency estimates based on higher order Gateaux derivatives and Taylor approximations to the design functional in order to show that the designs discussed in this paper are really close to the optimum. Furthermore we will discuss designs not only for prediction alone but for the estimation of the covariance function and a combined criterion for estimation and prediction, too. The approach will be similar to the approach followed in this paper. Strength and ideas can be borrowed again from experimental design theory.

### 4 Appendix: Convex design theory

#### 4.1 The experimental design problem for Bayesian linear regression

Here we give a survey of convex design theory, since, as is shown in Sections 1 and 2 of the paper, this theory can easily be used to solve spatial sampling design problems with correlated errors and to calculate minimax predictors. In contrast to the spatial sampling design problem the experimental design problem for linear regression models with uncorrelated errors may be considered as solved since the pioneering works of Kiefer (1959) and Fedorov (1972).

##### 4.1.1 The Bayesian linear regression model

To be specific, we consider here the following Bayesian linear regression model

$$Y(x) = \mathbf{h}(x)^T \gamma + \epsilon_0(x) \\ E(Y(x)) = \mathbf{h}(x)^T \gamma,$$

where  $\mathbf{h}(\mathbf{x})$  is a vector of regression functions in  $R^r$  dependent on explanatory variables  $x \in R^q, q \geq 1$ . We assume the error process  $\epsilon_0(x)$  to be uncorrelated with fixed variance

$$\text{var}(\epsilon_0(x)) = \sigma_0^2.$$

Further, we assume to have prior knowledge about the trend parameter vector  $\gamma$  in the form of a known prior mean  $E(\gamma) = \gamma_0$  and a known a priori covariance matrix  $\text{cov}(\gamma) = \mathbf{\Gamma} > 0$ . Then the best affine linear predictor



of the response surface at an unmeasured explanatory vector  $x_0$  is given by

$$\hat{Y}(x_0) = \mathbf{h}(x_0)^T \hat{\gamma}^B,$$

where

$$\hat{\gamma}^B = (\mathbf{H}^T \mathbf{H} + \sigma_0^2 \mathbf{\Gamma}^{-1})^{-1} (\mathbf{H}^T \mathbf{Y} + \sigma_0^2 \mathbf{\Gamma}^{-1} \gamma_0)$$

is the a posteriori mean of the trend parameter vector  $\gamma$  under the Gaussian assumption for the prior distribution and for the errors. Here  $\mathbf{H} = (\mathbf{h}(x_1), \dots, \mathbf{h}(x_n))^T$  is the design matrix and  $\mathbf{Y} = (Y(x_1), \dots, Y(x_n))^T$  is the vector of the observations.

#### 4.1.2 The Bayesian experimental design problem

The total mean squared error of our predictor reads

$$\begin{aligned} E((Y(x_0) - \hat{Y}(x_0))^2) &= \\ &= \sigma_0^2 [1 + \mathbf{h}(x_0)^T (\sigma_0^2 \mathbf{\Gamma}^{-1} + \mathbf{H}^T \mathbf{H})^{-1} \mathbf{h}(x_0)]. \end{aligned}$$

We write  $v_n = (x_1, x_2, \dots, x_n)$  for the exact design and  $\mathbf{H}(v_n) = \mathbf{H}$  to express, that the above expression is dependent on the selected design points  $x_i$ . Bayesian experimental design then tries to select the points in  $v_n$  in such a way, that the expression

$$\int_{\mathbf{X}} \mathbf{h}(x)^T (\mathbf{H}^T(v_n) \mathbf{H}(v_n) + \sigma_0^2 \mathbf{\Gamma}^{-1})^{-1} \mathbf{h}(x) dP(x)$$

is minimized, where  $\mathbf{X}$  is the design region and  $P(\cdot)$  is a fixed prespecified probability measure on  $\mathbf{X}$ , weighting the importance of the sampling points. Obviously, this is a discrete optimization problem and is therefore very complicated to solve numerically. The matrix

$$\mathbf{M}_B(v_n) = \frac{1}{n} (\mathbf{H}^T(v_n) \mathbf{H}(v_n) + \sigma_0^2 \mathbf{\Gamma}^{-1})$$

is called Bayesian information matrix. Its inverse is proportional to the expectation of the posterior covariance matrix of  $\gamma$ . Assuming that  $k$  different design points  $x_1, x_2, \dots, x_k$  with multiplicities  $n_1, n_2, \dots, n_k$ ,  $\sum_{i=1}^k n_i = n$  are contained in  $v_n$  we may write

$$\frac{1}{n} \mathbf{H}^T(v_n) \mathbf{H}(v_n) = \sum_{i=1}^k \frac{n_i}{n} \mathbf{h}(x_i) \mathbf{h}(x_i)^T.$$

The proportions  $\frac{n_i}{n}$  may be interpreted as probabilities or intensities with which the different design points  $x_i$  are selected. This interpretation is the key idea to define continuous design measures  $\xi(dx)$  on  $\mathbf{X}$ . A continuous design measure is just a probability measure on  $\mathbf{X}$ , and is a generalization of exact design measures for which

$$\begin{aligned} \xi(x_1) &= \frac{n_1}{n}, \xi(x_2) = \frac{n_2}{n}, \dots, \xi(x_k) = \frac{n_k}{n} \\ \sum_{i=1}^k n_i &= n, x_1 \neq x_2 \neq \dots \neq x_k, n_i \in \mathbb{N}. \end{aligned}$$

Then, we can define also a continuous Bayesian information matrix

$$\mathbf{M}_B(\xi) = \int_{\mathbf{X}} \mathbf{h}(x) \mathbf{h}(x)^T \xi(dx) + \frac{\sigma_0^2}{n} \mathbf{\Gamma}^{-1}$$

and a continuous Bayesian design problem

$$\int_{\mathbf{X}} \mathbf{h}(x)^T \mathbf{M}_B(\xi)^{-1} \mathbf{h}(x) P(dx) \rightarrow \min_{\xi \in \Xi}.$$

Here  $\Xi$  is the set of all probability measures on the set  $\mathbf{X}$ , which is supposed to be compact. Defining

$$\mathbf{U} = \int_{\mathbf{X}} \mathbf{h}(x) \mathbf{h}(x)^T P(dx),$$

the above minimization problem may be written as

$$\Psi(\xi) = \text{tr}(\mathbf{U} \mathbf{M}_B(\xi)^{-1}) \rightarrow \min_{\xi \in \Xi},$$

where  $\text{tr}(\cdot)$  is the trace functional. It may be shown that the set of all information matrices on  $\Xi$  is convex and compact. Furthermore it is possible to show that all functionals of the above form are convex and continuous in  $\mathbf{M}_B(\xi)$  and  $\xi$ . The above design functional  $\Psi(\xi)$  thus attains its minimum at a design  $\xi^* \in \Xi$ , see Pilz (1991).

#### 4.2 An algorithm for solving the continuous design problem

For the minimization of the design functional

$$\Psi(\xi) = \text{tr}(\mathbf{U} \mathbf{M}_B(\xi)^{-1})$$

algorithms may be used, as they are known from the theory of convex optimization. We concentrate here on an algorithm of Fedorov (1972). A more extensive exposure of design algorithms may be found in Pilz (1991). The proposed design algorithm is based on the directional derivative at  $\xi \in \Xi$  in the direction of  $\bar{\xi} \in \Xi$ :

$$\Delta_\psi(\xi, \bar{\xi}) = \lim_{\alpha \downarrow 0} \frac{d}{d\alpha} \Psi((1 - \alpha)\xi + \alpha\bar{\xi})$$

It may be shown that this directional derivative attains its minimum in the direction of a one-point design measure  $\delta_x \in \Xi$  with support  $x \in \mathbf{X}$ .

The proposed design algorithm is based on a fixed sequence  $\{\alpha_s\}_{s \in \mathbb{N}}$ , such that

$$\lim_{s \rightarrow \infty} \alpha_s = 0, \sum_{s \in \mathbb{N}} \alpha_s = \infty, \alpha_s \in [0, 1), s = 1, 2, \dots$$

At the  $s$ -th iteration of the design algorithm we determine

$$\Delta_\psi(\xi_{s-1}, \delta_{x_s}) = \inf_{x \in \mathbf{X}} \Delta_\psi(\xi_{s-1}, \delta_x)$$

and form the new design measure

$$\xi_s = (1 - \alpha_s) \xi_{s-1} + \alpha_s \delta_{x_s}.$$

It may be shown that the sequence  $\{\Psi(\xi_s)\}_{s \in \mathbb{N}}$  of functional values converges, and it holds

$$\lim_{s \rightarrow \infty} \Psi(\xi_s) = \inf_{\xi \in \Xi} \Psi(\xi).$$

Since  $\mathbf{M}_B(\xi)$  is regular whatever the design  $\xi \in \Xi$ , the starting design may be taken as one-point design  $\xi_0 = \delta_{x_0}$ , where  $x_0$  should be such that

$$\Psi(\xi_0) = \inf_{x \in \mathbf{X}} \Psi(\delta_x).$$

Then the weights  $\alpha_s$  needed for the construction of the  $s$ -th iteration design may be taken as  $\alpha_s = \frac{1}{1+s}$ ,  $s = 1, 2, \dots$ . The question remaining just is, when to stop iteration. The answer is dependent on the so-called efficiency of the design  $\xi$ :

$$e_\Psi(\xi) = \frac{\inf_{\xi \in \Xi} \Psi(\xi)}{\Psi(\xi)}.$$

Defining  $d_\Psi(\xi) = \inf_{x \in \mathbf{X}} \Delta_\Psi(\xi, \delta_x)$ , it may be shown that the following inequality holds for the efficiency:

$$1 + \frac{d_\Psi(\xi)}{\Psi(\xi)} \leq e_\Psi(\xi) \leq 1.$$

This way, the iteration should be stopped at stage  $s_0$  if

$$1 + \frac{d_\Psi(\xi_{s_0})}{\Psi(\xi_{s_0})} \geq e_0,$$

where  $e_0 \in (0, 1)$  is some predetermined efficiency that is to be guaranteed. We refer to Pilz (1991) for the explicit form of the necessary directional derivatives.

If the obtained discrete optimal design can be written as an exact design, i.e. if its weights are integer multiples of  $\frac{1}{n}$ , this design is also optimal within the class of exact designs. But in general this is not the case and a discrete design cannot be realized by observations. The obtained discrete design  $\xi^* = \{(x_1, p_1), \dots, (x_m, p_m)\}$  thus has to be rounded by means of forming the set

$$\Xi_n = \{\xi_{v_n} = \{(x_1, \frac{n_1}{n}), \dots, (x_m, \frac{n_m}{n})\} :$$

$$n_i \geq [np_i], n_i \in \{0, 1, \dots, n\}, \sum n_i = n\},$$

where  $[np_i]$  denotes the integer part of  $np_i$ . Hereafter we choose that design  $\xi_{v_n}^* \in \Xi_n$  as approximation to the optimal exact design for which  $\Psi(\xi_{v_n}^*)$  becomes a minimum in  $\Xi_n$ .

### 4.3 Iteration procedures for determining exact designs

We are now going to formulate iteration procedures for the construction of approximately optimal exact designs. Contrary to the construction of optimal discrete designs, here we cannot prove convergence of the exact designs to the functional value  $\Psi(v^*)$  of an optimal exact design  $v^*$ ; we can only guarantee stepwise

improvement of a given exact starting design, i.e. the sequence of functional values  $\Psi(v_{n,s})$  decreases monotonically with increasing iteration index  $s$ . The algorithm is an exchange type algorithm improving  $n$ -point designs and starting from an initial design.

#### 4.3.1 Exchange type algorithm

Step 1. Use some initial design  $v_{n,1} = (x_{1,1}, \dots, x_{n,1}) \in \mathbf{X}^n$  of size  $n$ .

Step 2. Beginning with  $s = 1$  form the design  $v_{n+1,s} = v_{n,s} + (x_{n+1,s})$  by adding the point

$$x_{n+1,s} = \arg \min_{x \in \mathbf{X}} \Psi(\mathbf{M}_B(v_{n,s} + (x)))$$

to  $v_{n,s}$ .

Then form  $v_{n,s}^j = v_{n+1,s} - (x_{j,s})$ ,  $j = 1, 2, \dots, n+1$  and delete that point  $x_{j^*,s}$  from  $v_{n+1,s}$  for which

$$\Psi(\mathbf{M}_B(v_{n,s}^j)) = \min_{j \in \{1, \dots, n+1\}} \Psi(\mathbf{M}_B(v_{n,s}^j)).$$

Step 3. Repeat Step 2 until the point to be deleted is equivalent to the point to be added.

For our design functional Step 2 is determined as follows:

$$x_{n+1,s} = \arg \max_{x \in \mathbf{X}} \frac{\mathbf{h}(x)^T \mathbf{M}_B(v_{n,s})^{-1} \mathbf{U} \mathbf{M}_B(v_{n,s})^{-1} \mathbf{h}(x)}{n + \mathbf{h}(x)^T \mathbf{M}_B(v_{n,s})^{-1} \mathbf{h}(x)}$$

$$j^* = \arg \min_{1 \leq j \leq n+1} \frac{\mathbf{h}(x_{j,s})^T \mathbf{Q}_B(v_{n+1,s}) \mathbf{h}(x_{j,s})}{n + 1 - \mathbf{h}(x_{j,s})^T \mathbf{M}_B(v_{n+1,s})^{-1} \mathbf{h}(x_{j,s})},$$

where

$$\mathbf{Q}_B(v_{n+1,s}) = \mathbf{M}_B(v_{n+1,s})^{-1} \mathbf{U} \mathbf{M}_B(v_{n+1,s})^{-1}.$$

#### 4.3.2 Generation of an initial design

The initial design is a one-point design which minimizes the design functional among all designs of size  $n = 1$ . Note that such a design exists since the Bayesian information matrix is positive definite even for designs of size  $n = 1$ .

Step 1. Choose  $x_1 \in \mathbf{X}$  such that

$$x_1 = \arg \min_{x \in \mathbf{X}} \Psi(\mathbf{M}_B((x))), \text{ and set } v_1 = (x_1).$$

Step 2. Beginning with  $i = 1$ , find  $x_{i+1}$  such that  $x_{i+1} = \arg \min_{x \in \mathbf{X}} \Psi(\mathbf{M}_B(v_i + (x)))$  and form  $v_{i+1} = v_i + (x_{i+1})$ . Continue with  $i$  replaced by  $i+1$  until  $i+1 = n$ .

Step 3. If  $i+1 = n$  then stop and take  $v_{n,1} = (x_1, \dots, x_n)$  as an initial design.

#### 4.3.3 Combination of the algorithms 4.3.2 and 4.3.1

It is a good idea to combine the initial design algorithm 4.3.2 and the exchange type algorithm 4.3.1 in the following way:

Step 1. Start with the initial design algorithm and find a design with one first design point.

Step 2. Having found a design with  $m \geq 1$  design points apply the exchange type algorithm to this design to improve it.

Step 3. Add to the design from Step 2 one further design point by means of the initial design algorithm to get  $m + 1$  design points.

Step 4. Go back to Step 2 and iterate Step 2 and Step 3 until you have found  $n$  desired design points.

#### 4.3.4 Reduction of experimental designs

Often it is desired to reduce a given experimental design  $v = (x_1, x_2, \dots, x_n)$  to one including only  $m < n$  design points from  $v$ :

Step 1. Delete that design point  $x_{j^*}$  from  $v$  for which  $x_{j^*} = \arg \min_{x_j \in v} \Psi(\mathbf{M}_B(v - (x_j)))$ , and set  $v := v - (x_{j^*})$ .

Step 2. Iterate Step 1 until the design  $v$  contains only  $m$  design points.

Also this algorithm may be combined with an improvement step similar to the exchange-type algorithm 4.3.1. In algorithm 4.3.1 merely the calculation of  $x_{n+1,s}$  has to be replaced by

$$x_{n+1,s} = \arg \min_{x \in v - v_{n,s}} \Psi(\mathbf{M}_B(v_{n,s} + (x))),$$

where  $v$  is the initial design that has to be reduced.

This improved algorithm has the advantage that design points once deleted can reenter the design in the exchange step.

#### 4.3.5 Exact design algorithms for the D-optimality criterion

Besides the Bayesian design criterion discussed up to now another well known convex design criterion is the so-called D-optimality criterion, which is equivalent to minimizing the determinant of the preposterior covariance matrix

$$(\sigma_0^2 \mathbf{\Gamma}^{-1} + \mathbf{H}(\mathbf{v}_n)^T \mathbf{H}(\mathbf{v}_n))^{-1}$$

of the regression parameter. The algorithms 4.3.1-4.3.4 may be applied to the D-optimality design criterion,

too, merely the formulae given for  $x_{n+1,s}$  and  $j^*$  change and become

$$x_{n+1,s} = \arg \max_{x \in \mathbf{X}} \mathbf{h}(x)^T \mathbf{M}_B(v_{n,s})^{-1} \mathbf{h}(x)$$

$$j^* = \arg \min_{1 \leq j \leq n+1} \mathbf{h}(x_{j,s})^T \mathbf{M}_B(v_{n+1,s})^{-1} \mathbf{h}(x_{j,s}).$$

#### 4.3.6 Inverse of the information matrix

Obviously the calculation of exact designs requires in every step the calculation of the inverses of the information matrices  $\mathbf{M}_B(v_{n,s})$  and  $\mathbf{M}_B(v_{n+1,s})$ . In the section on spatial sampling design we see that these information matrices can have a quite high dimension of about  $2500 \times 2500$ . So, how can one invert such large matrices in affordable time? A first artificial, inverse information matrix in spatial sampling design can always be one with block-diagonal structure corresponding to 0 selected design points, having one very small block, being the a priori covariance matrix for deterministic trend functions, and having one further block, being just a diagonal matrix of very high dimension (about 2500 diagonal elements, being the variances of the stochastic amplitudes resulting from a harmonic decomposition of the random field into sine-cosine-Bessel surface harmonics). So, no inversion is needed at a first step. The inversion of all other information matrices becomes easy, and there is computationally no need to make explicit use of numerical matrix inversion algorithms, when one considers equations (13.26) and (13.28) in Pilz (1991):

$$\begin{aligned} \mathbf{M}_B(v_{n,s} + (x))^{-1} &= \frac{n+1}{n} \{ \mathbf{M}_B(v_{n,s})^{-1} \\ &- \frac{\mathbf{M}_B(v_{n,s})^{-1} \mathbf{h}(x) \mathbf{h}(x)^T \mathbf{M}_B(v_{n,s})^{-1}}{n + \mathbf{h}(x)^T \mathbf{M}_B(v_{n,s})^{-1} \mathbf{h}(x)} \}, \\ \mathbf{M}_B(v_{n,s}^j)^{-1} &= \frac{n}{n+1} \{ \mathbf{M}_B(v_{n+1,s})^{-1} \\ &+ \frac{\mathbf{M}_B(v_{n+1,s})^{-1} \mathbf{h}(x_{j,s}) \mathbf{h}(x_{j,s})^T \mathbf{M}_B(v_{n+1,s})^{-1}}{n+1 - \mathbf{h}(x_{j,s})^T \mathbf{M}_B(v_{n+1,s})^{-1} \mathbf{h}(x_{j,s})} \} \end{aligned}$$

Obviously only matrix- and vector multiplications are needed in these update formulae.

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